

Integrating pfSense Firewall with Wazuh SIEM

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1 Introduction

A firewall is one of the most important security controls in modern computer networks. It monitors and filters incoming and outgoing network traffic based on predefined security rules, preventing unauthorized access while allowing legitimate communication.

Firewalls help organizations:

- Prevent unauthorized network access.
- Block malicious traffic.
- Enforce security policies.
- Monitor network connections.
- Protect internal systems from external threats.

Several types of firewalls exist, including packet-filtering firewalls, stateful inspection firewalls, proxy firewalls, and Next-Generation Firewalls (NGFWs), each providing different levels of traffic inspection and security.

In this project, **pfSense** was integrated with **Wazuh SIEM** to create a centralized monitoring environment. pfSense generated firewall and network security logs, while Wazuh collected, analyzed, and generated alerts based on those events, providing better visibility into network activities within the SOC laboratory.

2 Lab Environment

The firewall monitoring lab consisted of the following components:

Host Machine

- macOS (MacBook Air M2)

Virtualization Platform

- UTM

3 Why Integrate pfSense with Wazuh?

While Wazuh provides endpoint monitoring, pfSense offers network-level visibility.

Combining both platforms allows security analysts to monitor:

- Allowed network traffic
- Blocked connection
- Firewall rule matches
- Authentication attempts
- VPN activity
- DHCP events
- DNS events
- Network attacks

This creates a more complete Security Operations Center (SOC) environment where both endpoint and network events can be investigated from a single dashboard.

4 Installing pfSense

The pfSense firewall was deployed as a virtual machine within the laboratory environment.

The following hardware configuration was used:

Component	Configuration
RAM	2 GB
CPU	2 Cores
Storage	20 GB
Network Adapter 1	NAT (WAN)
Network Adapter 2	Internal LAN

After installation, the WAN and LAN interfaces were configured, allowing pfSense to function as the gateway between the internal lab network and the external network.

4.1 Select the Installation Media

Choose Installer disc image file (ISO) and browse for the downloaded pfSense ISO.

4.2 Name Your VM

Use a descriptive name such as pfSense and pick a save location.

4.3 Allocate Disk Space

- 20 GB minimum (store as a single file recommended).

4.4 Customize Hardware

- RAM: 1 GB minimum (2 GB recommended).
- CPU: 1 vCPU (2 cores recommended).
- Network Adapters:
 - Adapter 1 (WAN) → NAT (simulates internet access).
 - Adapter 2 (LAN) → Host-only (isolated internal lab network).

4.5 Assign Interfaces

- WAN (em0) → Configured via DHCP (IP in 192.x.x.x/24 range).
- LAN (em1) → Static IP 192.168.1.1/24.

```
*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: [REDACTED]/24
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Figure : pfSense interface

5 Accessing the pfSense Dashboard

After completing the installation, the pfSense web interface was accessed using a web browser.

Default credentials:

Username	Password
admin	pfsense

After the initial login, the default administrator password was changed to improve security.

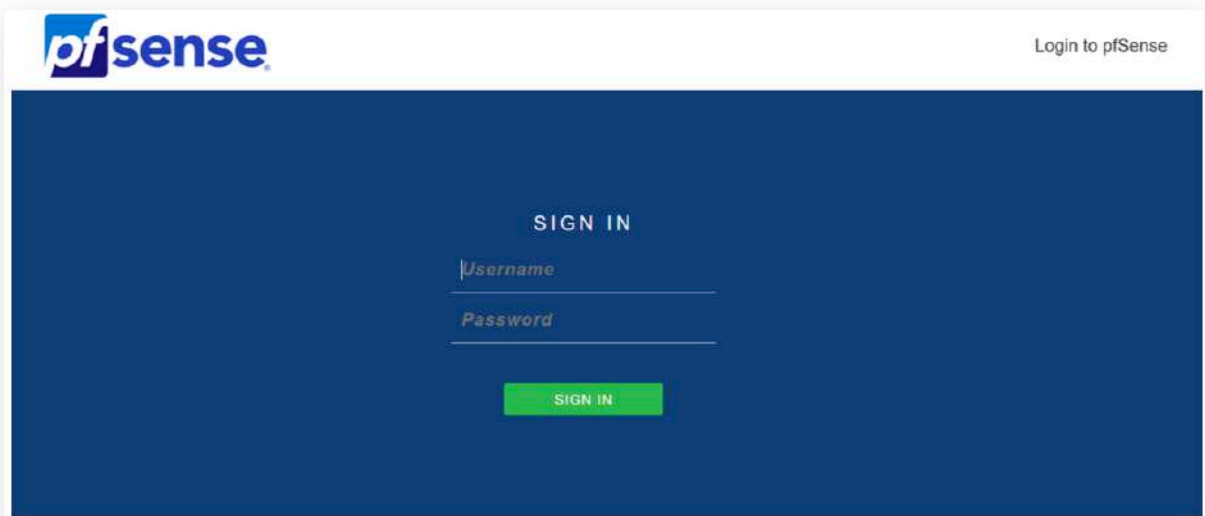


Figure : pfSense web interface login

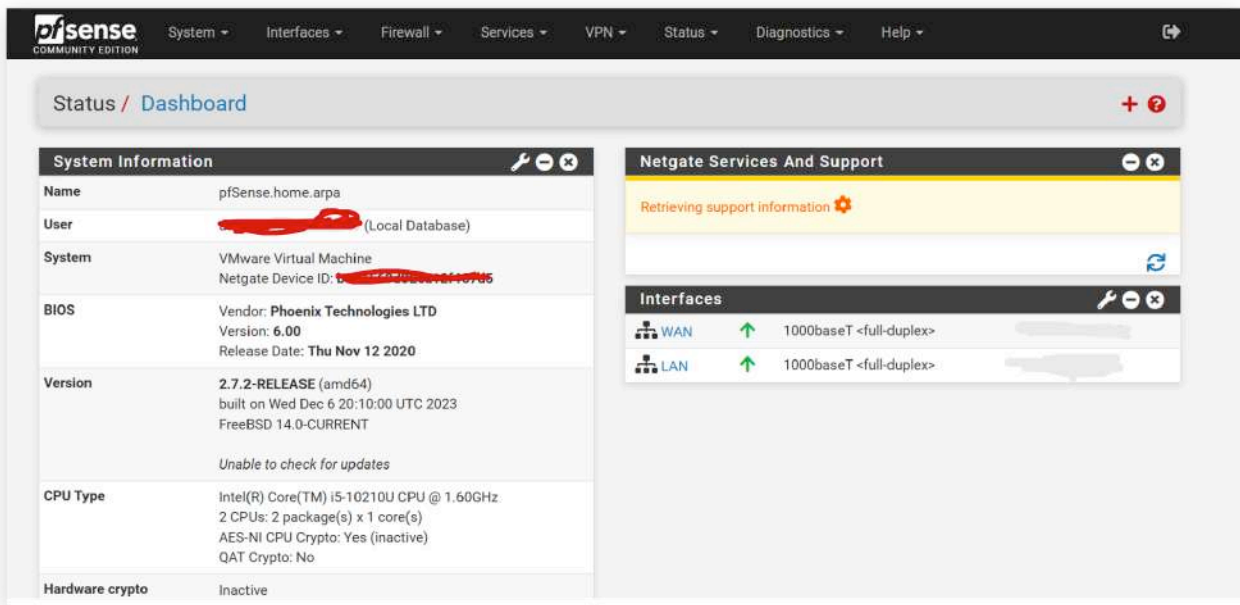


Figure : pfSense dashboard

6 Configuring Remote Logging

To forward firewall events to Wazuh, remote logging was enabled within pfSense.

Navigation:

Status → System Logs → Settings

The following configuration was applied:

- Enable Remote Logging
- Remote Syslog Server = Ubuntu Wazuh Server IP
- Protocol = UDP
- Port = 514

The following log categories were enabled:

- Firewall
- System
- DHCP
- DNS Resolver
- VPN (if configured)

After saving the configuration, pfSense began forwarding firewall logs to the Wazuh server.



The screenshot shows the 'Remote Logging Options' configuration page in pfSense. The 'Enable Remote Logging' checkbox is checked, and the 'Send log messages to remote syslog server' checkbox is also checked. The 'Source Address' is set to 'Default (any)'. The 'IP Protocol' is set to 'IPv4'. The 'Remote log servers' field shows '192.168.1.1:514' and 'IP[:port]'. The 'Remote Syslog Contents' section has 'Everything' checked, and several other categories are listed with checkboxes: System Events, Firewall Events, DNS Events (Resolver/unbound, Forwarder/dnsmasq, filterdns), DHCP Events (DHCP Daemon, DHCP Relay, DHCP Client), and PPP Events (PPPoE WAN Client, L2TP WAN Client, PPTP WAN Client).

Figure : pfSense remote logging configuration

6.1 Configure Wazuh Syslog Input

On the Wazuh Manager, edit `/var/ossec/etc/ossec.cfg`:

```
<ossec_config>

<!-- pfSense syslog input -->

<remote>

<connection> syslog </connection>

<port> 514 </port>

<protocol> udp </protocol>

<allowed-ips> YOUR_PF_SENSE_IP </allowed-ips>

<local_ip> YOUR_WAZUH_SERVER_IP </local_ip>

</remote>

</ossec_config>
```

7 Configuring Wazuh to Receive pfSense Logs

The Wazuh Manager was configured to receive syslog events from the pfSense firewall.

The Wazuh configuration file was updated to enable remote syslog collection.

The configuration specified:

- Syslog connection
- UDP protocol
- Listening port (514)
- Allowed pfSense IP address
- Local Wazuh Manager IP address

After saving the configuration, the Wazuh Manager service was restarted.

Once restarted, the manager began receiving syslog events from pfSense.

8 Creating Custom Decoders

pfSense firewall logs are generated using the **filterlog** format, which is not fully parsed by Wazuh's default decoders.

To correctly extract useful information from firewall events, custom decoders were created.

The decoder extracted important fields such as:

- Source IP Address
- Destination IP Address
- Protocol
- Source Port
- Destination Port
- Firewall Action

This allowed Wazuh to interpret pfSense logs correctly and generate meaningful alerts.

A decoder extracts key fields (source IP, destination IP, protocol).

Example (pfsense-custom-decoder.xml):

```
< decoder name = " pfsense - custom " >
< prematch > filterlog </ prematch >
</ decoder >

< decoder name = " pfsense - fields " >
< parent > pfsense - custom </ parent >

< regex > (\ w +) \ d + : \ S * , \ S * , \ S * , ( \ S +) , \ S * , \ S * , ( \ S +) , \ S * , \ S * , \ S
* , \ S * , \ S * , \ S * , \ S * , \ S * , \ S * , \ S * , ( \ S +) , \ S * , ( \ S +) , ( \ S +) , ( \ d
+) ; ( \ d +) , \ S * </ regex >

< order > logsource , id , section , protocol , srcip , dstip , srcport ,
dstport </ order >
</ decoder >
```


9 Creating Custom Decoder & Rules

Since pfSense logs have a unique format, Wazuh may not parse them correctly out of the box.

9.1 Custom Decoder

A decoder extracts key fields (source IP, destination IP, protocol).

Example (pfsense-custom-decoder.xml):

```
< decoder name =" pfsense - custom " >
< prematch > filterlog </ prematch >
</ decoder >
< decoder name =" pfsense - fields " >
< parent > pfsense - custom </ parent >
< regex >(\ w +) \ d +: \ S * , \ S * , \ S * , ( \ S +) , \ S * , \ S * , ( \ S +) , \ S * , \ S * , \ S
    * , \ S * , \ S * , \ S * , \ S * , \ S * , \ S * , \ S * , ( \ S +) , \ S * , ( \ S +) , ( \ S +) , ( \ d
    +) ; ( \ d +) , \ S * </ regex >
< order > logsource , id , section , protocol , srcip , dstip , srcport ,
dstport </ order >
</ decoder >
```

9.2 Custom Rules

Custom Wazuh rules were created to classify pfSense firewall events according to their severity.

A rule assigns severity levels and alerts based on matched events.

Example (pfsense-custom-rules.xml):

```
< group name = " pfsense , firewall , " >
  <! -- Generic pfSense log -- >
  < rule id = " 100111 " level = " 3 " >
    < decoded_as > pfsense - custom </ decoded_as >
    < description > pfSense log received </ description >
  </ rule >

  <! -- Successful login -- >
  < rule id = " 100112 " level = " 3 " >
    < if_sid > 100111 </ if_sid >
    < match > Successful login </ match >
    < description > pfSense: Successful login from $( srcip ) </
      description >
  </ rule >

  <! -- Allowed traffic -- >
  < rule id = " 100113 " level = " 5 " >
    < if_sid > 100111 </ if_sid >
    < match > pass </ match >
    < description > pfSense: Allowed traffic from $( srcip ) to $(
      dstip ) </ description >
  </ rule >

  <! -- Blocked traffic -- >
  < rule id = " 100114 " level = " 7 " >
    < if_sid > 100111 </ if_sid >
    < match > block </ match >
    < description > pfSense: Blocked traffic from $( srcip ) to $(
      dstip ) </ description >
  </ rule >

  <! -- Authentication error -- >
  < rule id = " 100115 " level = " 10 " >
    < if_sid > 100111 </ if_sid >
    < match > authentication error </ match >
    < description > pfSense: Authentication error for user $(
      dstuser ) from $( srcip ) </ description >
  </ rule >
</ group >
```

The following rule categories were implemented:

Rule Type	Severity
Successful Login	Level 3
Allowed Traffic	Level 5
Blocked Traffic	Level 7
Authentication Error	Level 10

These custom rules enabled Wazuh to generate alerts whenever matching firewall events were received.

A decoder extracts key fields (source IP, destination IP, protocol).

Example (pfsense-custom-decoder.xml):

10 Viewing pfSense Events in Wazuh

After the integration was completed, firewall events became visible inside the Wazuh Dashboard.

Navigate to:

Threat Hunting

The dashboard displayed events such as:

- Allowed firewall traffic
- Blocked firewall traffic
- Successful administrator logins
- Authentication failures
- DHCP events
- DNS requests
- VPN events (if enabled)

Each event included valuable information for investigation, including:

- Timestamp
- Source IP
- Destination IP
- Protocol
- Source Port
- Destination Port
- Firewall Action
- Rule Level
- Alert Description

These logs allowed firewall activity to be correlated with endpoint events collected from Windows systems.

11 Benefits of Firewall Log Correlation

By integrating pfSense with Wazuh, both endpoint and network activities could be analyzed together.

Examples include:

- Correlating blocked firewall traffic with failed SSH login attempts.
- Investigating suspicious external IP addresses.
- Monitoring network scanning activities.
- Detecting repeated authentication failures.
- Tracking allowed and denied network connections.

This centralized visibility improves threat detection and accelerates incident response.

12 Conclusion

This project successfully integrated the pfSense firewall with the Wazuh SIEM platform to create a centralized network security monitoring solution. Firewall logs generated by pfSense were forwarded to the Wazuh Manager using remote syslog, where they were parsed using custom decoders and analyzed through custom detection rules.

The integration enabled real-time monitoring of firewall events such as allowed connections, blocked traffic, authentication attempts, and other network activities. By combining pfSense firewall logs with endpoint telemetry collected by the Wazuh Agent, the SOC laboratory achieved improved visibility across both network and host systems.

This integration provides a strong foundation for future enhancements such as Suricata IDS/IPS integration, VirusTotal lookups, automated active response, and advanced threat correlation within the Wazuh platform.